

F5 Exponential equations

Name: _____

Class: _____

Date: _____

Time: **154 minutes**

Marks: **127 marks**

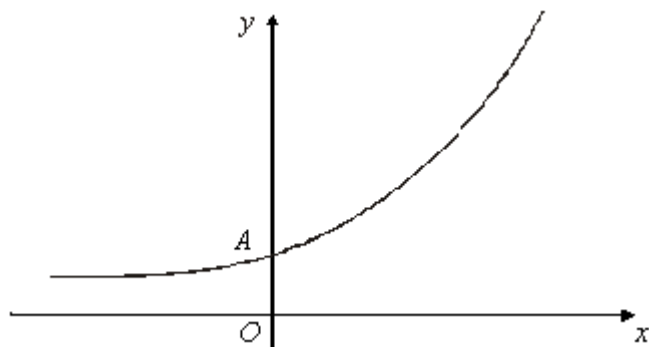
Comments:

EXAM PAPERS PRACTICE

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Q1.

The diagram shows a sketch of the curve with equation $y = 3(2^x + 1)$.



The curve $y = 3(2^x + 1)$ intersects the y -axis at the point A .

- (a) Find the y -coordinate of the point A . (2)
 - (b) Use the trapezium rule with four ordinates (three strips) to find an approximate value for $\int_0^6 3(2^x + 1) dx$ (4)
 - (c) The line $y = 21$ intersects the curve $y = 3(2^x + 1)$ at the point P .
 - (i) Show that the x -coordinate of P satisfies the equation $2^x = 6$ (1)
 - (ii) Use logarithms to find the x -coordinate of P , giving your answer to three significant figures. (3)
- (Total 10 marks)**

Q2.

A biologist is researching the growth of a certain species of hamster. She proposes that the length, x cm, of a hamster t days after its birth is given by

$$x = 15 - 12e^{-\frac{t}{14}}$$

- (a) Use this model to find:
 - (i) the length of a hamster when it is born; (1)

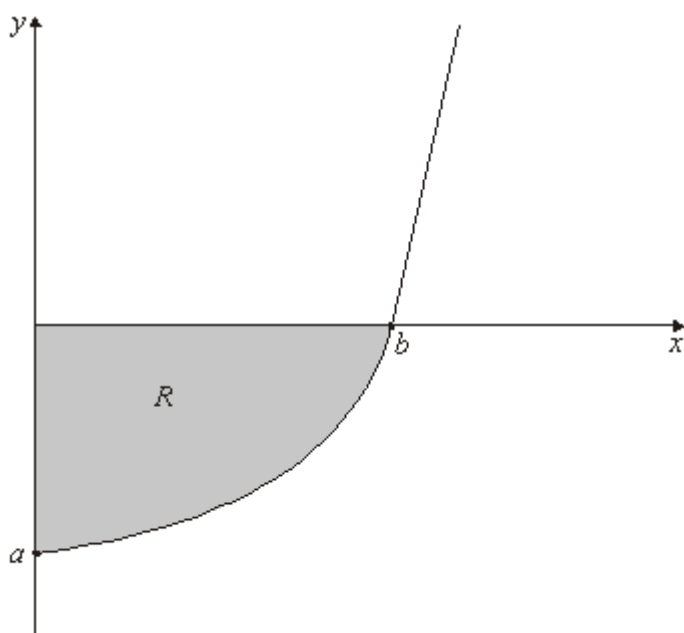
- (ii) the length of a hamster after 14 days, giving your answer to three significant figures. (2)
- (b) (i) Show that the time for a hamster to grow to 10 cm in length is given by $t = 14 \ln \left(\frac{a}{b} \right)$, where a and b are integers. (3)
- (ii) Find this time to the nearest day. (1)
- (c) (i) Show that $\frac{dx}{dt} = \frac{1}{14}(15 - x)$ (3)
- (ii) Find the rate of growth of the hamster, in cm per day, when its length is 8 cm. (1)
- (Total 11 marks)**

Q3.

- (a) Use logarithms to solve the equation $0.8^x = 0.05$, giving your answer to three decimal places. (3)
- (b) An infinite geometric series has common ratio r . The sum to infinity of the series is five times the first term of the series.
- (i) Show that $r = 0.8$. (3)
- (ii) Given that the first term of the series is 20, find the least value of n such that the n th term of the series is less than 1. (3)
- (Total 9 marks)**

Q4.

The diagram shows part of the graph of $y = e^{2x} - 9$. The graph cuts the coordinate axes at $(0, a)$ and $(b, 0)$.



- (a) State the value of a , and show that $b = \ln 3$.

(3)

- (b) Show that $y^2 = e^{4x} - 18e^{2x} + 81$.

(1)

(Total 4 marks)

Q5.

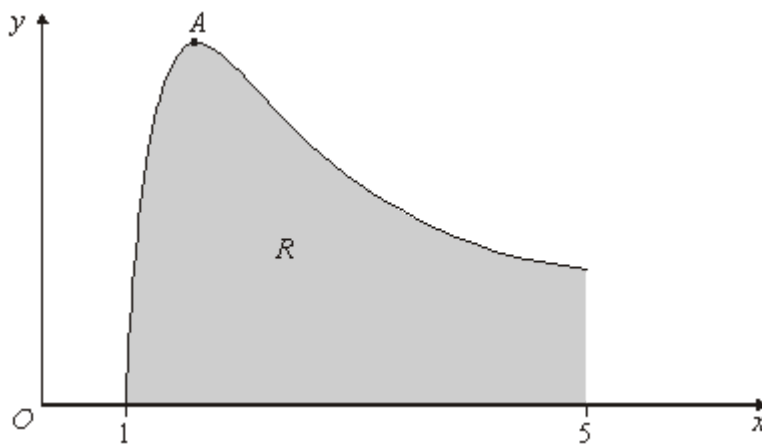
- (a) Given that $y = x^{-2} \ln x$, show that $\frac{dy}{dx} = \frac{1 - 2 \ln x}{x^3}$.

(4)

- (b) Using integration by parts, find $\int x^{-2} \ln x \, dx$.

(4)

- (c) The sketch shows the graph of $y = x^{-2} \ln x$.



- (i) Using the answer to part (a), find, in terms of e , the x -coordinate of the stationary point A . (2)
- (ii) The region R is bounded by the curve, the x -axis and the line $x = 5$. Using your answer to part (b), show that the area of R is

$$\frac{1}{5} (4 - \ln 5)$$

(3)

(Total 13 marks)

Q6.

On 1 January 1900, a sculpture was valued at £80.

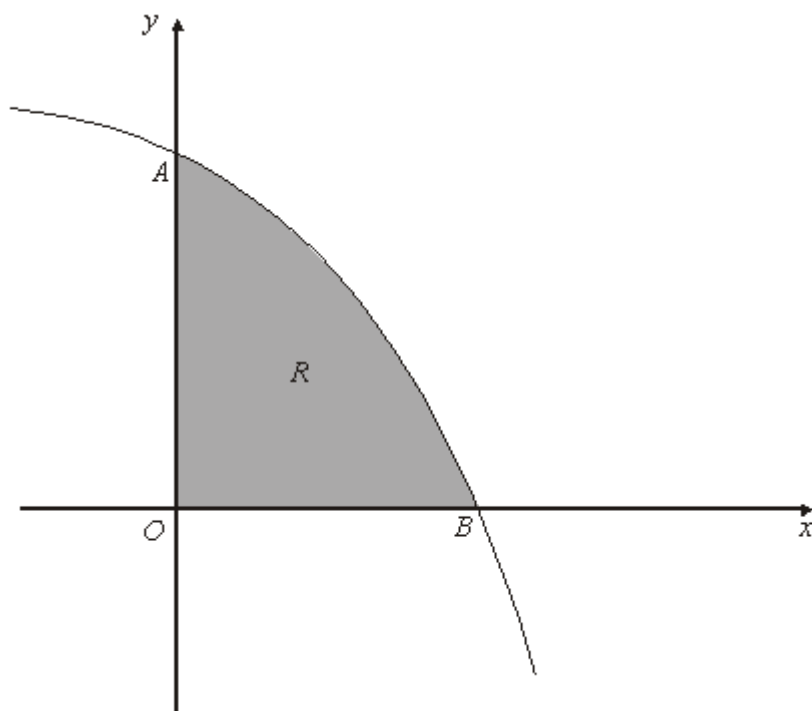
When the sculpture was sold on 1 January 1956, its value was £5000.

The value, £ V , of the sculpture is modelled by the formula $V = A k^t$, where t is the time in years since 1 January 1900 and A and k are constants.

- (a) Write down the value of A . (1)
- (b) Show that $k \approx 1.07664$. (3)
- (c) Use this model to:
- (i) show that the value of the sculpture on 1 January 2006 will be greater than £200 000; (2)
- (ii) find the year in which the value of the sculpture will first exceed £800 000. (3)
- (Total 9 marks)

Q7.

The diagram shows a sketch of the curve with equation $y = 27 - 3^x$.



The curve $y = 27 - 3^x$ intersects the y -axis at the point A and the x -axis at the point B .

- (a) (i) Find the y -coordinate of point A . (2)
- (ii) Verify that the x -coordinate of point B is 3. (1)
- (b) The region, R , bounded by the curve $y = 27 - 3^x$ and the coordinate axes is shaded. Use the trapezium rule with four ordinates (three strips) to find an approximate value for the area of R . (4)
- (c) (i) Use logarithms to solve the equation $3^x = 13$, giving your answer to four decimal places. (3)
- (ii) The line $y = k$ intersects the curve $y = 27 - 3^x$ at the point where $3^x = 13$.
Find the value of k . (1)
- (d) (i) Describe the single geometrical transformation by which the curve with equation $y = -3^x$ can be obtained **from** the curve $y = 27 - 3^x$. (2)
- (ii) Sketch the curve $y = -3^x$. (2)

(Total 15 marks)

Q8.

(a) A curve has equation $y = e^{2x} - 10e^x + 12x$.

(i) Find $\frac{dy}{dx}$.

(2)

(ii) Find $\frac{d^2y}{dx^2}$.

(1)

(b) The points P and Q are the stationary points of the curve.

(i) Show that the x -coordinates of P and Q are given by the solutions of the equation

$$e^{2x} - 5e^x + 6 = 0$$

(1)

(ii) By using the substitution $z = e^x$, or otherwise, show that the x -coordinates of P and Q are $\ln 2$ and $\ln 3$.

(3)

(iii) Find the y -coordinates of P and Q , giving each of your answers in the form $m + 12 \ln n$, where m and n are integers.

(3)

(iv) Using the answer to part (a)(ii), determine the nature of each stationary point.

(3)

(Total 13 marks)

Q9.

A disease is spreading through a colony of rabbits. There are 5000 rabbits in the colony. At time t hours, x is the number of rabbits infected. The rate of increase of the number of rabbits infected is proportional to the product of the number of rabbits infected and the number not yet infected.

(a) (i) Formulate a differential equation for $\frac{dx}{dt}$ in terms of the variables x and t and a constant of proportionality k .

(2)

(ii) Initially, 1000 rabbits are infected and the disease is spreading at a rate of 200 rabbits per hour. Find the value of the constant k .

(You are **not** required to solve your differential equation.)

(2)

- (b) The solution of the differential equation in this model is

$$t = 4 \ln \left(\frac{4x}{5000 - x} \right)$$

- (i) Find the time after which 2500 rabbits will be infected, giving your answer in hours to one decimal place.
- (ii) Find, according to this model, the number of rabbits infected after 30 hours.

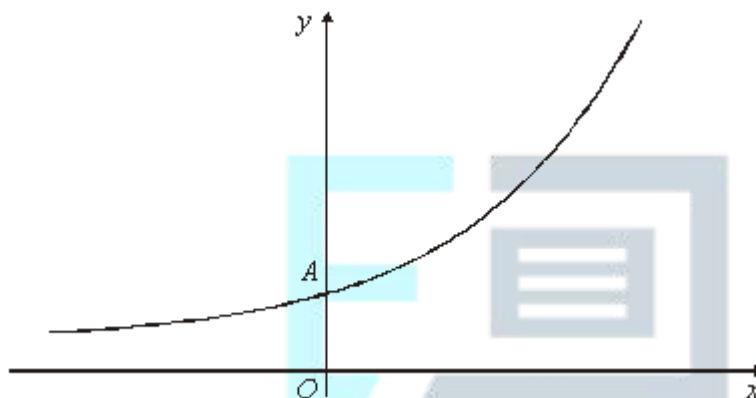
(2)

(4)

(Total 10 marks)

Q10.

The diagram shows a sketch of the curve with equation $y = 3^x + 1$.



The curve intersects the y -axis at the point A .

- (a) Write down the y -coordinate of point A .
- (b) (i) Use the trapezium rule with five ordinates (four strips) to find an approximation for

$$\int_0^1 (3^x + 1) dx$$

, giving your answer to three significant figures.

(1)

(4)

- (ii) By considering the graph of $y = 3^x + 1$, explain with the aid of a diagram whether your approximation will be an overestimate or an underestimate of the true value of

$$\int_0^1 (3^x + 1) dx$$

(2)

- (c) The line $y = 5$ intersects the curve $y = 3^x + 1$ at the point P . By solving a suitable equation, find the x -coordinate of the point P . Give your answer to four decimal places.

(4)

- (d) The curve $y = 3^x + 1$ is reflected in the y -axis to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$.

(1)

(Total 12 marks)

Q11.

- (a) Solve the equation $2e^x = 5$, giving your answer as an exact natural logarithm.

(2)

- (b) (i) By substituting $y = e^x$, show that the equation $2e^x + 5e^{-x} = 7$ can be written as

$$2y^2 - 7y + 5 = 0$$

(2)

- (ii) Hence solve the equation $2e^x + 5e^{-x} = 7$, giving your answers as exact values of x .

(3)

(Total 7 marks)

Q12.

- (a) A cup of coffee is cooling down in a room. At time t minutes after the coffee is made, its temperature is x °C, where

$$x = 15 + 70e^{-\frac{t}{40}}$$

- (i) Find the temperature of the coffee when it is made.

(1)

- (ii) Find the temperature of the coffee 30 minutes after it is made.

(2)

- (iii) Find how long it will take for the coffee to cool down to 60 °C.

(3)

- (b) (i) Use integration to solve the differential equation

$$\frac{dx}{dt} = -\frac{1}{40}(x-15), \quad x > 15$$

given that $x = 85$ when $t = 0$, expressing t in terms of x .

(6)

- (ii) Hence show $x = 15 + 70e^{-\frac{t}{40}}$

(2)

(Total 14 marks)